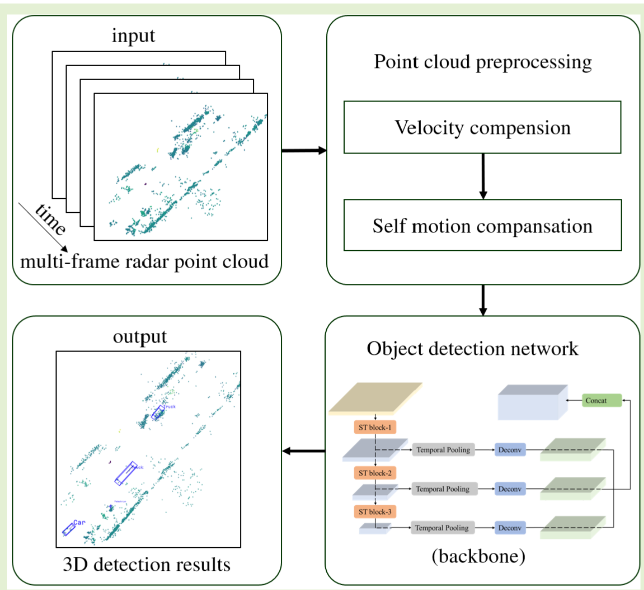
IEEE Sensors Council IEEE SENSORS JOURNAL, VOL. 23, NO. 11, 1 JUNE 2023

3-D Object Detection for Multiframe 4-D Automotive Millimeter-Wave Radar Point Cloud

Bin Tan, Zhixiong Ma, Xichan Zhu, Sen Li, Lianqing Zheng, Sihan Chen, Libo Huang, and Jie Bai

***Abstract*—Object detection is a crucial task in autonomous driving. Currently, object-detection methods for autonomous driving systems are primarily based on information from cameras and light detection and ranging (LiDAR), which may experience interference from complex lighting or poor weather. At present, the 4-D (x, y, z, v) millimeter-wave radar can provide a denser point cloud to achieve 3-D object-detection tasks that are difficult to complete with traditional millimeter-wave radar. Existing 3-D object point-cloud-detection algorithms are mostly based on 3-D LiDAR; these methods are not necessarily applicable to millimeter-wave radars, which have sparser data and more noise and include velocity information. This study proposes a 3-D object-detection framework based on a multiframe 4-D millimeter-wave radar point cloud. First, the ego vehicle velocity information is estimated by the millimeter-wave radar, and the relative velocity information of the millimeter-wave radar point cloud is compensated for the absolute velocity. Second, by matching between millimeter-wave radar frames, the multiframe millimeter-wave radar point cloud is matched to the last frame. Finally, the object is detected by the proposed multiframe millimeter-wave radar point-cloud-detection network. Experiments are performed using our newly recorded TJ4DRadSet dataset in a complex traffic environment. The results showed that the proposed object-detection framework outperformed the comparison methods based on the 3-D mean average precision. The experimental results and methods can be used as the baseline for other multiframe 4-D millimeter-wave radar-detection algorithms.**



***Index Terms*—Autonomous driving, deep learning, millimeter-wave radar, object detection, point clouds.**

I. Introduction

OBJECT detection is an essential part of autonomous vehicle environment perception. 3-D object detection obtains object category information and geometric information, such as length, width, and height. In an autonomous driving scenario, the vehicle must conduct subsequent planning and control according to the 3-D object-detection information. At present, vehicles primarily use monocular cameras, binocular cameras, multiline light detection and ranging (LiDAR), and millimeter-wave radar to carry out 3-D object detection [1].

Monocular and binocular cameras can output rich object image features. They are widely used for 2-D and 3-D object detection; however, their applicability is limited under complex environmental conditions, such as rainy and foggy weather, and strong light exposure. LiDAR has a high range and angle resolution, and it can output dense point-cloud information and provide 3-D scene information. Therefore, LiDAR is widely used in 3-D object detection; however, it is expensive and affected by weather conditions.

Traditional 2+1D (x, y, v) millimeter-wave radar can effectively measure the radial distance, radial velocity, and horizontal-angle information of objects. The 2+1D radar can work under complex environmental conditions. However, it cannot detect the object type, and obtaining specific size and orientation information is difficult. Compared with LiDAR, millimeter-wave radar has limited range and angular resolution, and the output points are too sparse; thus, it is

Manuscript received 22 September 2022; accepted 24 October 2022. Date of publication 9 November 2022; date of current version 31 May 2023. This work was supported by the National Key Research and Development Program of China under Grant 2021YFB2501201. The associate editor coordinating the review of this article and approving it for publication was Dr. Guofa Li. *(Corresponding authors: Zhixiong Ma; Jie Bai.)*

Bin Tan, Zhixiong Ma, Xichan Zhu, Sen Li, Lianqing Zheng, and Sihan Chen are with the School of Automotive Studies, Tongji University, Shanghai 201804, China (e-mail: tanbin@tongji.edu.cn; mzx1978@tongji.edu.cn; zhuxichan@tongji.edu.cn; lisen@tongji.edu.cn; zhenglianqing@tongji.edu.cn; sihan.chen@tongji.edu.cn).

Libo Huang and Jie Bai are with the School of Information and Electricity, Zhejiang University City College, Hangzhou, Zhejiang 310015, China (e-mail: huangl@zucc.edu.cn; baij@zucc.edu.cn).

Digital Object Identifier 10.1109/JSEN.2022.3219643

generally used in the final object-fusion stage of autonomous driving perception. Moreover, it is difficult to perform the 3-D detection of objects, such as vehicles and pedestrians, based on radar points.

The development of radar technology produced high-resolution 4-D radar, which can provide additional height information and detect more characteristic object information, such as object-edge information.

At present, object detection based on millimeter-wave radar point clouds is still undeveloped, and compared with LiDAR data, the distribution of millimeter-wave radar point-cloud information is sparse. When designing a deep learning network, it is necessary to consider the object radial-velocity information output by the 4-D radar [2], [3]. However, the sparse point cloud of single-frame millimeter-wave radar may induce problems, such as a difficult regression of the object heading angle and contour in 3-D object detection. Therefore, this study makes a novel contribution to the literature by proposing a multiframe millimeter-wave radar point-cloud-detection framework that can accumulate multiframe information for object detection.

The overall framework consists of three parts. First, the ego vehicle velocity information is estimated by the millimeter-wave radar, and the relative-speed information of the millimeter-wave radar point cloud is compensated for the absolute speed. Second, by matching between millimeter-wave radar frames, the multiframe millimeter-wave radar point cloud is matched to the last frame. Finally, the targets are detected by the proposed multiframe millimeter-wave radar point-cloud-detection network. The network primarily consists of three modules: the pseudoimage encoder, the backbone extractor, and the detection head. The first module encodes the multiframe millimeter-wave radar point cloud into a unified tensor with spatiotemporal information, and the other two forms learn to extract the spatiotemporal information of the target and fuse the spatiotemporal information at multiple scales of the target.

In summary, an efficient object-detection method based entirely on the multiframe 4-D radar point cloud is proposed. Our main contributions are given as follows.

1) We propose a complete object-detection framework based on a multiframe millimeter-wave radar point cloud that consists of multiple parts, including radar point-cloud velocity compensation, radar interframe matching, and multiframe point-cloud detection. Moreover, the velocity information of the target is utilized in the matching between radar frames.
2) A spatiotemporal network is constructed, named RadarMFNet, and it is based on millimeter-wave radar point-cloud object detection and fusing the spatiotemporal information at multiple spatial scales in the spatiotemporal feature extraction.
3) Different single- and multi-frame-based 3-D object-detection methods are compared using the dataset, and our method achieves the best 3-D mean average precision (mAP) results. Most of the existing 3-D object-detection methods are based on LiDAR; however, the experimental results show that these methods do not provide adequate detection when based on 4-D radar. In an actual complex road environment, a 4-D radar point cloud provides a better-detection effect on pedestrian and cyclist objects; however, the detection effect on car and truck objects is still limited.

II. Related Work

A. Millimeter-Wave Radar Point-Cloud-Based Object Detection

Traditional 2+1D (x, y, v) millimeter-wave radar outputs the radial distance, the azimuth angle, the intensity, and the radial velocity of objects. At present, there are a variety of research methods for object detection using traditional millimeter-wave radar point clouds, including traditional machine learning methods [2], [3], [4] and deep learning methods [5], [6].

4-D radar can output the elevation angle of an object, and its azimuth angular resolution is improved by dense multiple-input–multiple-output antenna channels. At present, the public datasets that contain 4-D radar data detection include the Astyx [7] and Delft datasets [8]. The Astyx dataset is small (500 frames), and the objects it contains are mainly vehicle objects. Related object-detection literature includes 3-D car detection via the combination of radar and cameras with the AVOD fusion network [9]. In addition, RPFA-Net detects vehicles using a novel pillar feature attention network [10]. The Delft dataset contains more than 8000 frames of synchronized sensor data, and the object types include vehicle, pedestrian, and cyclist objects. In a previous study, 3-D object detection was performed on radar point clouds via pointpillar radar, which is a pointpillar network trained on all five radar features [8]. We built an autonomous driving dataset called TJ4DRadSet [11], which consists of more than 7000 frames of multimodal sensors, including 4-D radar, LiDAR, and cameras.

B. LiDAR Point-Cloud-Based Object Detection

Many LiDAR single-frame-detection deep learning methods can be divided into grid-based and point-set-based methods [12], [13].

The grid-based methods convert 3-D space into a regular grid so that it can be processed by 3-D and 2-D convolutional neural networks (CNNs). Some classic methods are given as follows. Voxelnet [14] converts 3-D space into a regular voxel, the features of the points in the voxel are extracted through the pointnet, and a 3-D CNN is then used to perceive a larger range. SECOND [15] uses sparse convolution to increase the speed of the network. Pointpillars [16] use pillars to encode the points' information in the voxel. PVRCNN [17] combines point- and voxel-based methods for feature extraction.

The point-set-based methods extract the local features near each point and then perform network calculations on these local features to output 3-D object information. Some classic methods are given as follows. 3-D SSD [18] is a pioneering method in this field. PointRCNN [19] generates possible proposals directly from the 3-D point cloud instead of using the projection method. STD [20] proposes a strategy to transform sparse data into dense data to obtain better results.

There are also various methods for 3-D object detection using multiple frames, including recurrent neural networks and

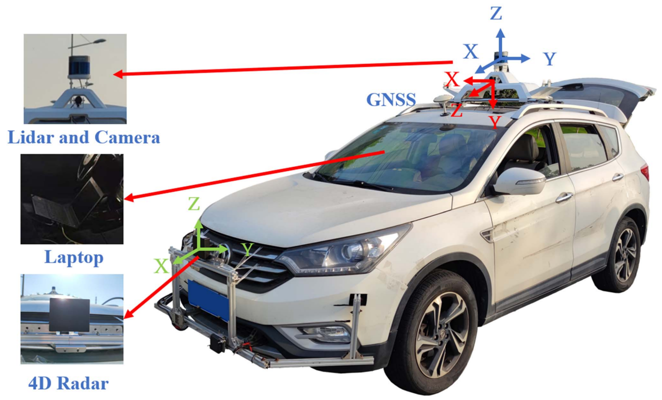


Fig. 1. Data acquisition platform, including 4-D radar, LiDAR, and camera sensor.

their variants [21], [22], graph neural networks [23], and 3-D CNNs [24].

C. Use of Millimeter-Wave Radar Doppler

Millimeter-wave radar can directly output the Doppler information of the target, which is the relative radial velocity. A target with the same absolute velocity has a different radial velocity at different positions. The Doppler velocity of millimeter-wave radar has been utilized in a number of ways. It can be used to distinguish between static and dynamic objects [25], [26]. It can be combined with the spatial information of the target point to estimate the state of the ego vehicle [27], [28], and it can also be used for simultaneous localization and mapping [29], [30]. The Doppler velocity can also be used in the research of millimeter-wave target classification, and it can be used to output target-velocity characteristics [2], [3]. In millimeter-wave object classification and detection, the addition of the radar Doppler velocity can enhance object classification and detection performance [6]. Using the fusion of millimeter-wave radar and LiDAR, RadarNet [31] predicts the movement direction of the radar target by aligning the radial velocity of the radar target with the detected movement direction. The use of millimeter-wave radar velocity information in fusion detection with some millimeter-wave radars with other sensors requires further research.

III. Dataset Acquisition

A. System Setup

The data acquisition platform primarily consists of 4-D radar, LiDAR, and camera sensors. As shown in Fig. 1, the 4-D radar is installed in the middle of the front ventilation grille, 0.5 m above the ground. The LiDAR can collect 360° of environmental information, and the camera and 4-D radar can capture information in the field of view (FOV), which includes the driving conditions ahead. The inertial measurement unit and the global positioning system receiving modules are fixed in the trunk of the vehicle, and the antenna is installed on the roof of the vehicle to receive high-precision differential signals. The main parameters of the sensors used in the whole data acquisition platform are shown in Table I.

TABLE I
Parameters of the Collection Sensor

Sensors	Resolution			FOV		
	Range	Azimuth	Elevation	Range	Azimuth	Elevation
Camera	/	1280px	960px	/	66.5°	94°
Lidar	0.03 m	0.1° - 0.4°	0.33°	120 m	360°	40°
4D radar	0.86 m	<1°	<1°	400 m	113°	45°

Multiple data-acquisition sensors are calibrated. This process mainly includes internal parameter calibration, external parameter calibration, and time-synchronization calibration. The camera's internal parameters and distortion coefficients were calibrated using a MATLAB (MathWorks, USA) toolkit and checkerboard. The internal parameters of the 4-D radar and LiDAR were calibrated when the sensors were manufactured. External parameter calibration was conducted for the camera and LiDAR, as well as the 4-D radar and LiDAR. The extrinsic parameters between the sensors are represented as translation and rotation matrices, and the ROS system is used to provide timestamps from multiple sensors.

B. Dataset

The dataset called TJ4DRadSet, which includes 7757 frames within 44 consecutive sequences, was collected in Suzhou, China, in the fourth quarter of 2021. Part of the collected data from the 4-D radar and camera is shown in Fig. 2, and some statistical information of the dataset is shown in Fig. 3. Various road scenes, lighting conditions at different time periods, and different objects are fully considered in the data-collection process. The collection of scenes includes typical urban and suburban areas, elevated highways, parks, industrial areas, and other simple scenes that include one-way and multilane roads, as well as densely targeted intersections and T-junctions. The collection period covers early morning, noon, evening, and night, and the collection includes both strongly backlit and low-light scenes. The collection objects include pedestrians; cyclists, such as bicycles and motorcycles; vehicles, such as passenger cars, buses, school buses, and sightseeing cars; and engineering and freight vehicles, such as trucks, cranes, and tankers. Some of the above scenarios and objects are unique to China. The acquisition platform is based on the ROS system, and the raw data of all sensors is recorded with a total size of approximately 527 GB. The ground-truth annotations of the dataset primarily include the 3-D bounding box, type, and trajectory identity of each object. The point-cloud density of LiDAR is higher than that of 4-D radar; thus, it can describe the shape of objects more clearly. Therefore, the annotation system mainly relies on the LiDAR point cloud and images for joint annotation. The 3-D bounding box of each object includes the center point (x, y, z), length, width, height (l, w, h), and orientation angle (yaw). In addition, the occlusion and truncation information has been annotated. The dataset is,

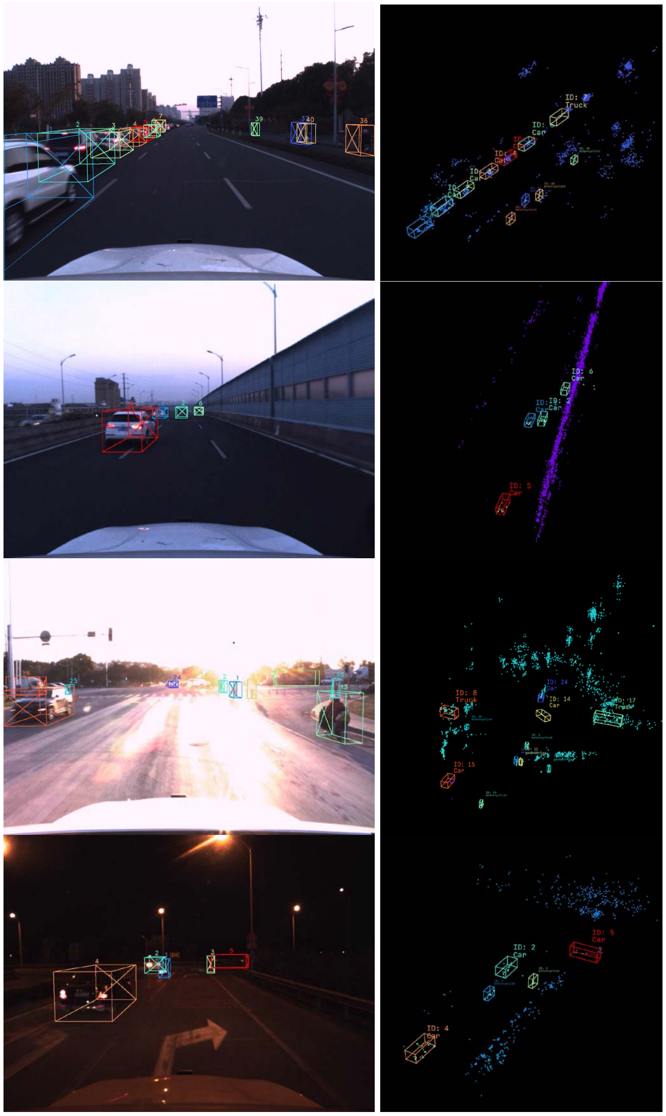


Fig. 2. Visualization results for typical scenarios from the dataset. The dataset includes scenes of different road environments, such as roads with nonmotorized vehicle lanes, two-way lanes, strongly lit environments, intersections, and dark night environments. (a) Camera data. (b) 4-D radar data.

finally, mapped into four classes (car, pedestrian, cyclist, and truck).

IV. Method

There are many differences in the data characteristics of 4-D radar point clouds and LiDAR point clouds. First, the point-cloud density distribution is different. The LiDAR point cloud has a high point-cloud density and can clearly present the shape of the object, whereas the 4-D radar point cloud has obvious sparsity, and sometimes, it is difficult to distinguish the shape and type of objects at the same distance. Compared with the single-frame millimeter-wave radar point cloud, the 3-D object-detection algorithm based on the multiframe millimeter-wave radar point cloud can better regress information, such as the orientation and shape of the target. Second, the 4-D radar point cloud can obtain the Doppler velocity and

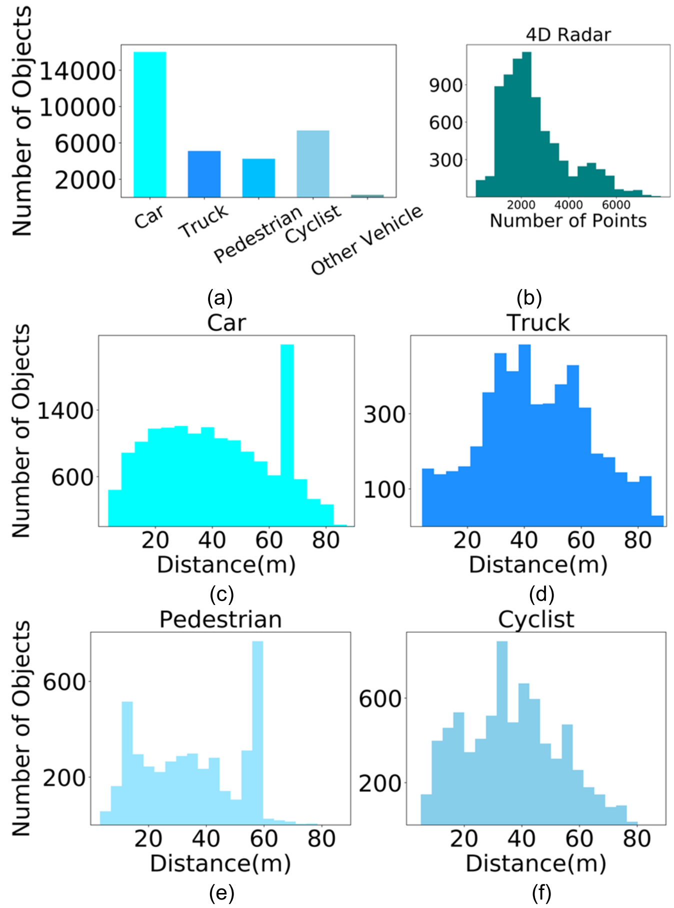


Fig. 3. Statistical properties of some data in the dataset. (a) Number of objects in each class. (b) 4-D radar point-cloud density. (c) Distribution of the distances of “Car”. (d) Distribution of the distances of “Truck”. (e) Distribution of the distances of “Pedestrian”. (f) Distribution of the distances of “Cyclist”.

radar reflection intensity, which represents the characteristics of the object. Moreover, the object-detection network proposed in this study also considers the target speed and the feature encoding of the reflection intensity.

The proposed millimeter-wave radar point-cloud data-processing flowchart is shown in Fig. 4.

A. Point-Cloud Preprocessing

The preprocessing method that is conducted before feeding the multiframe millimeter-wave radar point cloud into the object-detection network is given as follows.

The direct output of the millimeter-wave radar is the relative radial velocity information of the target point. Thus, before the multiframe millimeter-wave radar point cloud is input to the object-detection network, the relative radial velocity of the millimeter-wave radar point cloud is compensated to the absolute radial velocity to reduce the influence of relative velocity changes. The absolute radial velocity can easily distinguish a dynamic target from a static target. Moreover, when classifying a dynamic target, the velocity distribution can be used as an essential feature of the target classification.

The world coordinate systems of the multiframe point clouds are different, owing to the movement of the ego vehicle. Multiframe millimeter-wave radar point clouds were

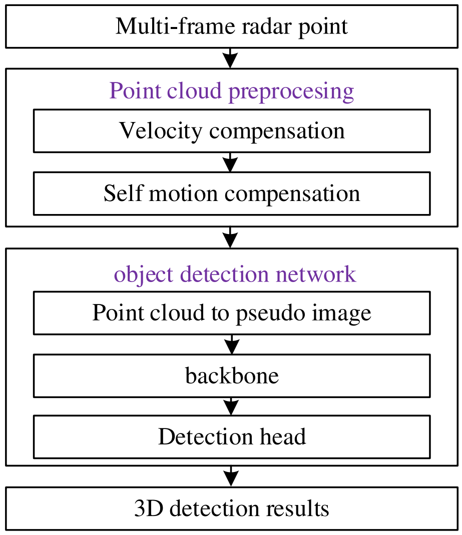


Fig. 4. Multiframe 4-D automotive millimeter-wave radar point-cloud data-processing flowchart, including point-cloud preprocessing and the object-detection network.

matched, and the point-cloud data at different vehicle positions were integrated into the last frame of the millimeter-wave radar coordinate system through rigid transformations, such as rotation and translation. The radar point-cloud self-motion compensation can reduce the impact of the vehicle's motion on the network regression.

1) Radar Point-Cloud Velocity Compensation: There are various methods that may be utilized to estimate the velocity of the ego vehicle at the millimeter-wave radar [39], [40].

1) The velocity and the angular velocity of the centroid of the ego vehicle are obtained using high-precision positioning information, and the vehicle velocity at the 4-D radar is then obtained using the external parameter matrix of the 4-D radar relative to the vehicle centroid.

2) The ego vehicle velocity at the 4-D radar is estimated using the information of the 4-D radar point cloud. We propose a simple approach based on method 2).

The radial velocity v_d^r of the point cloud measured by the 4-D radar can be expressed as the dot product of the point-cloud direction and target velocity v_d

$$v_d^r = v_{d,x}\frac{x}{R} + v_{d,y}\frac{y}{R} + v_{d,z}\frac{z}{R} \tag{1}$$

where R is the radial distance of the target.

As shown in Fig. 5, all radar static-object points have an equal relative velocity v_s, which is opposite to the velocity of the ego vehicle v_c at the position of the 4-D radar sensor. Given a set of N measurements, the above expression is applied

$$\underbrace{\begin{bmatrix} v_{s,1}^r \\ v_{s,2}^r \\ \cdots \\ v_{s,N}^r \end{bmatrix} = \begin{bmatrix} \frac{x_1}{R_1} & \frac{y_1}{R_1} & \frac{z_1}{R_1} \\ \frac{x_2}{R_2} & \frac{y_2}{R_2} & \frac{z_2}{R_2} \\ \cdots & \cdots & \cdots \\ \frac{x_N}{R_N} & \frac{y_N}{R_N} & \frac{z_N}{R_N} \end{bmatrix} \begin{bmatrix} v_{s,x} \\ v_{s,y} \\ v_{s,z} \end{bmatrix}}_{v_s = H v^r}. \tag{2}$$

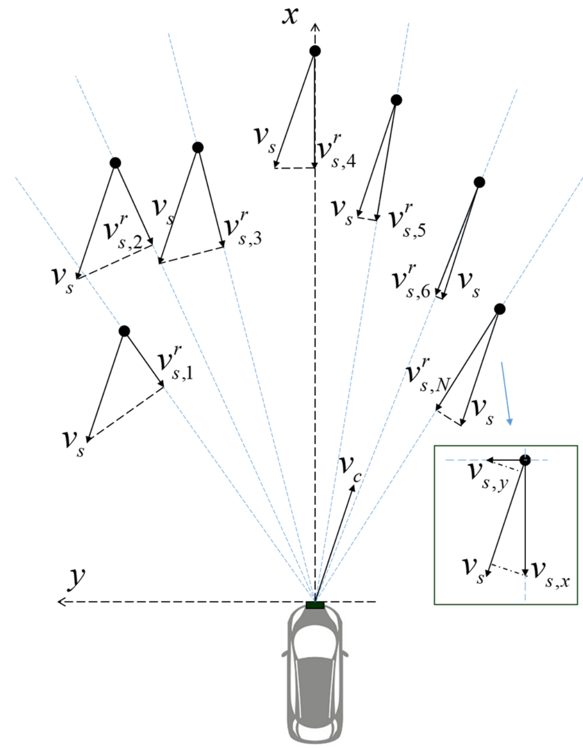


Fig. 5. Relationship between the radial velocity v_s^r, target static velocity v_s, and ego vehicle velocity v_c. The schematic shows the top view of the scene.

Equation (2) can be solved using the linear least-squares method

$$\hat{v}_s = \left(H^T H\right)^{-1} H^T v_s^r. \tag{3}$$

From (3), the velocity of the ego vehicle $\hat{v}_c$ is estimated based on information from the radar static-object points. There are errors to consider when solving the above formula using all measurements because the points measured by the radar are not all static points; therefore, the above formula is solved using the random sample and consensus method [32]. This model is used to fit experimental data, and it is able to smooth data that contain a particular amount of error. In the model proposed in this study, these errors are caused by moving objects, which will produce a skewed velocity distribution.

After solving the ego vehicle speed $\hat{v}_c$, the radial velocity compensation $\hat{v}_c^r$ of the target can be calculated using (1), and the speed of each target point v_a^r can be compensated as

$$v_a^r = v_d^r - \hat{v}_c^r. \tag{4}$$

2) Radar-Point-Cloud Self-Motion Compensation: Ego-motion compensation is performed on the point cloud via multiframe point-cloud matching to reduce the influence of vehicle ego-motion on the detection effect of the network.

For the nth frame point cloud of the radar, the point set P_n^m can be expressed as

$$P_n^m = \left\{p_n^1, p_n^2, \ldots, p_n^m\right\} \tag{5}$$

$$p_n^m = \left[x_n^m, y_n^m, z_n^m\right] \tag{6}$$

where p_n^m is the information of the mth point, and x_n^m, y_n^m, z_n^m are the xyz-coordinate information of the mth point.

The absolute velocity of the radar point cloud is obtained using (4), and the static point cloud can be obtained by filtering the velocity threshold. The static point set Q_n^l is expressed as

$$Q_n^l = \left\{ q_{_1}^1, q_{_1}^2, \ldots, q_{_n}^l \right\} \tag{7}$$

where $q_{_n}^l$ is the information of the lth point.

For point-cloud matching between radar frames, we need to determine the Euclidean transformation R, t, such that

$$\forall i, \quad q_n^i = R_{n-1} q_{n-1}^i + t_{n-1} \tag{8}$$

where R is the rotation matrix and t is the translation matrix.

The above transformation matrix is solved using the iterative closest point (ICP) algorithm [33]. This algorithm performs iterative calculations based on the least-squares method such that the sum of the squares of the errors reaches a minimum value

$$\arg\min_{R,t} \sum_{i=1}^{l} \left\| \left(R_{n-1} q_{n-1}^i + t_{n-1} \right) - q_n^i \right\|^2. \tag{9}$$

The radar point-cloud conversion relationship can be expressed as

$$H = [R, t] \tag{10}$$

$$Y_{n-1}^n = H_{n-1} P_{n-1}. \tag{11}$$

Y_{n-1}^n is the point set after the point cloud of the $(n-1)$th frame is registered to the point cloud of the nth frame.

For the point set after the $(n-m)$th frame converted to the nth frame point cloud

$$Y_{n-m}^n = H_{n-1} Y_{n-m}^{n-1}. \tag{12}$$

In multiframe radar detection, the nth frame and the first m frames after registration are input to the network in the nth frame

$$I_n = \left[\, P_n \; Y_{n-1}^n \; \ldots \; Y_{n-m}^n \, \right]. \tag{13}$$

From (12) and (13), the ICP algorithm is calculated only once in each frame iteration to obtain H_{n-1}, and it is then calculated from the point set of the previous frame I_{n-1}.

B. Object-Detection Network

The proposed multiframe 4-D radar point-cloud-detection network RadarMFNet consists of three processes: 1) multiframe point-cloud feature encoding and pseudoimage generation; 2) a spatiotemporal backbone network; and 3) detection head regression. Point-cloud coding is the process of extracting multiframe point-cloud features, and it also processes irregular point-cloud input into the same data format according to particular rules. The spatiotemporal backbone network is primarily used for feature extraction, and a 3-D CNN is mainly used to process spatiotemporal pseudoimage grid data. The detection head module primarily performs regression-task detection on the 3-D box position, orientation, and type of the object. The specific network structure is shown in Fig. 6.

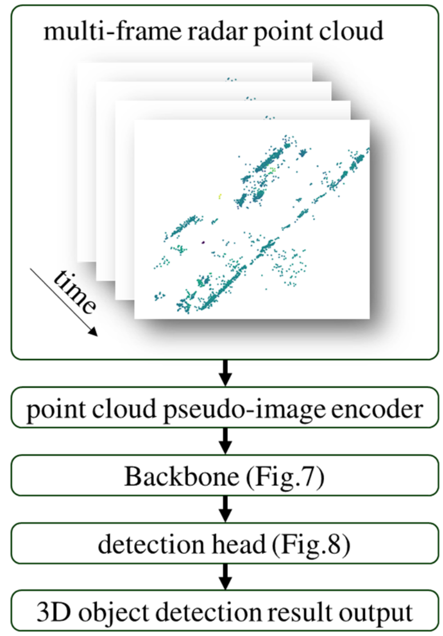


Fig. 6. Multiframe 4-D automotive millimeter-wave radar point-cloud data-processing flowchart, including point-cloud pseudoimage generation, spatiotemporal feature-extraction backbone network, and single-stage-detection head.

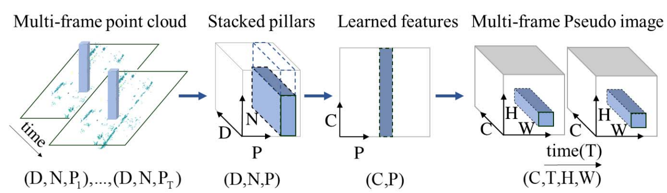


Fig. 7. Multiframe point-cloud pseudoimage-generation process.

1) Point Cloud to Pseudoimage: At present, there are two main point-cloud coding methods. The first is the point-based method, which uses a ball query to conduct point-cloud sampling, and its representative network is PointRCNN [30]. It has a flexible receptive field range and better detection results for small objects. However, large-scale point-cloud computing is inefficient using this method, and objects are required to have good point-cloud distribution characteristics. The second is the Voxel-based method, which transforms a large range of a point cloud into a tensor of a fixed-size grid combination, and its representative networks are VoxelNet [25] and SECOND [26]. Experiments have shown that the sparsity of the point-cloud distribution of millimeter-wave radar makes it challenging to extract object features using the point-based coding method because the distribution of millimeter-wave radar points contains a greater amount of noise. Therefore, the voxel-based method was used to encode the point cloud. In addition, the encoding of the height direction was ignored, and pillar-based encoding was used to reduce the amount of computation. Our method is shown in Fig. 7.

First, the original 4-D radar point cloud of each frame of the multiframe 4-D radar point cloud is cropped to the range of $R_x \times R_y \times R_z$ along the X-, Y-, and Z-axes. Using the original information of each radar point cloud, the dimension is expanded to generate the enhancement point l_i

$$l_i = \left[x, y, z, v, I, x_c, y_c, z_c, v_c, I_c, x_p, y_p, z_p\right] \tag{14}$$

where $[x, y, z, v, I]$ represents the original measurement coordinates of the radar point cloud, Doppler velocity, and reflection-intensity information, respectively. c represents the arithmetic mean distance of each point cloud with respect to all points in the pillar, and p represents the distance from the point cloud to the center offset of the pillar. Thus, each enhancement point is described by a $D = 13$ dimensional vector. Finally, a dense tensor of size (D, P_i, N) can be obtained for each frame point cloud, where P is the number of nonempty pillars, N is the number of points per pillar, and i represents the ith frame.

The tensors of size (D, P, N) are obtained by combining the point clouds of T frames. Then, the points in the D dimension are operated on, and tensors of size (C, P, N) are obtained through the linear layer, the batch-normalization (BN) layer, and the ReLU layer consecutively. Then, maxpooling is performed on the point-cloud quantity dimension to obtain an output tensor of size (C, P). The encoded features are dispersed to the original pillar position, and finally, a pseudoimage of size (C, T, H, W) is obtained, where C represents the number of channels, T represents the number of frames, and H and W are the height and width of the pseudoimage, respectively.

2) Backbone: The backbone network in the detection stage is used to extract temporal and spatial features, and the network structure is shown in Fig. 8.

The backbone network primarily consists of two subnetworks: a top-down subnetwork extracts features at the same temporal resolution at decreasing spatial resolutions, and the other network performs top-down feature upsampling and concatenation.

The top-down subnetwork can be represented by a series of ST blocks: the stride of the first convolution in the layer is 2, and the result of the first layer is obtained by adding $1 \times 3 \times 3$ and $3 \times 3 \times 3$ convolutions and dividing by 2. The stride of the subsequent convolution in the module is 1, which consists of L $1 \times 3 \times 3$ convolutions. Each convolution in the module is followed by BN and ReLU layers.

The output features of each top-down module are combined by temporal pooling, upsampling, and concatenation. First, the 4-D tensor (C, T, H, W) is converted into a 3-D tensor (C, H, W) without the time dimension via max-pooling at the time dimension. A 2-D transposed convolution is then used to upsample the feature map with an initial stride S_{in} and final stride S_{out}. Then, the features are sequentially upsampled through the BN and ReLU layers; the final output result is the concatenation result of features from different spatial resolutions. Because 3-D convolution is used in the top-down module, it is possible to extract the spatiotemporal features of the target.

3) Detection Head: The single-stage object-detector heads for 3-D object detection, which are similar to detection networks, such as pointpillar and SECOND, were used in this study.

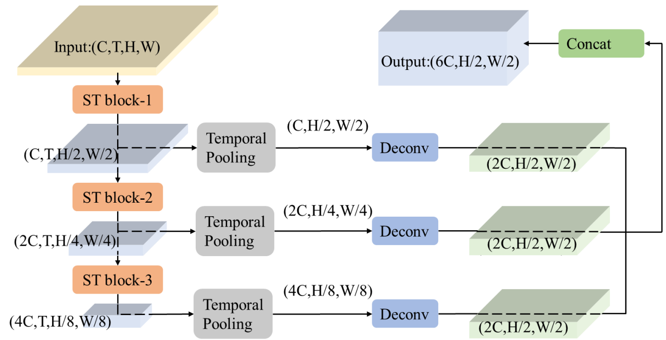


Fig. 8. Backbone network for spatiotemporal feature extraction, which includes a top-down feature-extraction network and an upsampling layer for multiscale feature fusion.

The ground-truth boxes and anchors can be defined by the set of parameters $(x, y, z, w, l, h, \theta)$. The residual of the ground-truth boxes and anchors can be defined as

$$\Delta x = \frac{x^{\text{gt}} - x^a}{d^a}, \quad \Delta y = \frac{y^{\text{gt}} - y^a}{d^a}, \quad \Delta z = \frac{z^{\text{gt}} - z^a}{h^a} \tag{15}$$

$$\Delta w = \log \frac{w^{\text{gt}}}{w^a}, \quad \Delta l = \log \frac{l^{\text{gt}}}{l^a}, \quad \Delta h = \log \frac{h^{\text{gt}}}{h^a} \tag{16}$$

$$\Delta\theta = \theta^{\text{gt}} - \theta^a \tag{17}$$

where the subscripts gt and a represent the ground-truth boxes and anchors, respectively, and $d^a = ((w^a)^2 + (l^a)^2)^{1/2}$.

The angle loss regression can be expressed as

$$L_\theta = \text{SmoothL1}\left(\sin\left(\theta_p - \Delta\theta\right)\right). \tag{18}$$

The total localization loss L_{loc} can be expressed as

$$L_{\text{loc}} = \sum_{b \in (x,y,z,w,l,h)} \text{SmoothL1}\left(b_p - \Delta b\right) + L_\theta \tag{19}$$

where b_p represents the regression value.

The SmoothL function is defined as

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise.} \end{cases} \tag{20}$$

The angle loss cannot distinguish the flipped boxes; therefore, a direction loss l_{dir} is used, which enables the network to learn the direction loss.

For the object classification loss L_{cls}, the focal loss has been used as follows:

$$L_{\text{cls}} = -\alpha_a \left(1 - p^a\right)^\gamma \log p^a \tag{21}$$

where p^a is the category probability of an anchor. Furthermore, α is 0.25, γ is 2, and the total loss L is

$$L = \frac{1}{N_{\text{pos}}} \left(\beta_{\text{loc}} L_{\text{loc}} + \beta_{\text{cls}} L_{\text{cls}} + \beta_{\text{dir}} L_{\text{dir}}\right) \tag{22}$$

where N_{pos} is the positive sample anchor, β_{loc} is 2, β_{cls} is 1, and β_{dir} is 0.2. The Adam optimization is used for the optimization of the loss function, and 80 epochs are trained with a batchsize of 4.

C. Implementation Details

1) Data Augmentation: To augment the dataset, different point-cloud data augmentation methods are employed. When the network inputs multiframe point clouds, the following data-enhancement processing methods are performed on each frame of the point clouds.

Random World Flip: Each ground-truth box and the point cloud are randomly flipped with the x-axis.

Random World Rotation: Each ground-truth box and the point cloud are randomly rotated by [−pi/4, pi/4].

Random World Scaling: Each ground-truth box and the point cloud are randomly scaled by [0.95, 1.05].

2) Network Settings: During point-cloud encoding and pseudoimage generation, all 3-D point-cloud spaces are cropped to $R_x = [0, 69.12]$, $R_y = [-39.68, 39.68]$, and $R_z = [-4, 2]$, and the ground-truth object boxes outside the area are removed. The point-cloud data are shuffled during training and not during testing. The point-cloud data are encoded into a pillar, and the pillar is 0.16 m in the x- and y-directions and 6 m in the z-direction; therefore, the maximum number of nonempty pillars in the space is $P = 214\,272$. In addition, the maximum number of point clouds in the pillar is $N = 32$. The current cloud is sampled when it exceeds the number N, and if it is insufficient, it is filled with 0 values.

In the backbone network, the input dimension C of the model is 64. As shown in Fig. 8, each top-down network consists of three ST blocks, the L value of subsequent convolutional layers in the three ST modules is (3, 5, 5), and the number of input and output channels of the modules can be calculated from the figure. As shown in Fig. 9, the first layer of the ST module is the downsampled layer, and the stride is 2. Then, the upsampling module contains three deconvolution modules, and the input and output dimensions of the modules can be calculated from Fig. 9. The features of the upsampling module are concatenated together to obtain the 6C-dimensional features of the detection head.

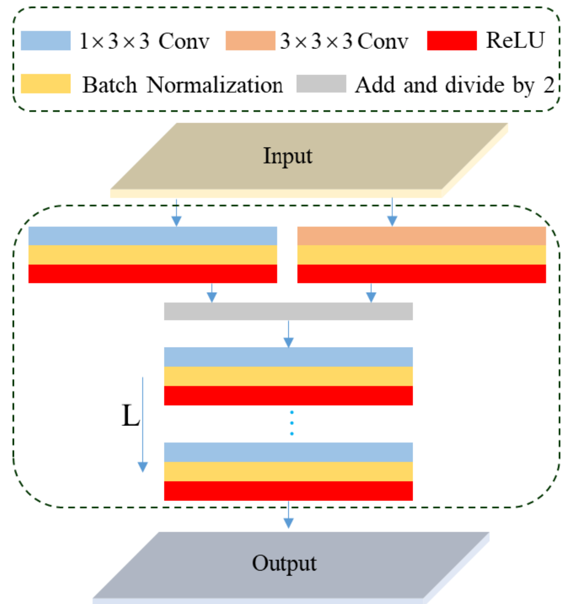


Fig. 9. Structure of the ST block, which consists of a series of convolution, BN, and ReLU layers. The first layer is obtained by adding 1 × 3 × 3 and 3 × 3 × 3 convolutions and dividing by 2. The first layer reduces the scale and increases the feature channel, and the subsequent layers include L 1 × 3 × 3 convolutions and further extract spatial features. Each convolution is followed by BN and ReLU layers.

V. Experiments

A. Experimental Settings

Using the collected dataset TJ4DRadSet, this study performs an algorithm performance evaluation and experimental comparison based on 3-D object detection. The dataset contains 7757 annotated 4-D radar point clouds within 44 consecutive sequences, which can be used for research, such as the 3-D detection of different objects. The training data are divided into a training set of 5717 frames and a test set of 2040 frames in a 7:3 schedule according to the scene.

For anchor-box matching, anchor boxes for each class are described by their width, length, height, and z-center, and they are applied to both the 0° and 90° orientations. In the pregenerated anchor box, the anchor dimension format is defined as (l, w, h). The anchor dimensions for "Car," "Truck," "Pedestrian," and "Cyclist" are given as follows: (4.56 m, 1.84 m, 1.70 m), (10.76 m, 2.66 m, 3.47 m), (0.80 m, 0.60 m, 1.69 m), and (1.77 m, 0.78 m, 1.60 m). The object intersection over union (IOU) threshold is 0.5 for the car and truck categories and 0.25 for the pedestrian and cyclist objects.

All object types are included in the experiment, and the average precision (AP) and mAP are used as evaluation metrics for 3-D detection. The AP is calculated using 40 recall positions, and the targets with zero points in the ground-truth box are filtered out during the precision calculation.

There are few algorithms for 3-D object detection in 4-D radar sparse point clouds; therefore, this experiment uses the performance of a point-cloud-detection-network model based on LiDAR for comparison. The input to all the algorithms in Table II is the 5-D information of the 4-D radar point cloud. The detection results include 3-D detection and 2-D bird's eye view (BEV) detection. Sections V-B and V-C analyzes the detection performance of the algorithm based on the quantitative and qualitative experimental results.

B. Qualitative Analysis

Fig. 10 visualizes the detection results of the model in different scenarios in the test set. The 3-D detection result in the picture is the true value, the blue box in the point-cloud-detection result is the 3-D true value box, and the green box is the predicted value.

In Fig. 10(a)–(c), it can be observed that both vehicle targets and pedestrians can obtain good detection results, and the network can distinguish the target contours from other objects, such as noise and the background. In some cases, as shown in Fig. 10(f), it is difficult to extract the target because the vehicle is a static target, and it is very close to the surrounding static objects, such as green belts; therefore, it is missed by

TABLE II
COMPARISON OF DETECTION RESULTS WITH DIFFERENT 3-D OBJECT-DETECTION METHODS#

	Method	3D AP					BEV AP					Params (M)	Inference time (ms)[d]
		Car	Pedestrian	Cyclist	Truck	mAp	Car	Pedestrian	Cyclist	Truck	mAp		
Single frame	3DSSD [18]	0.08	4.84	11.25	0.22	4.10	0.36	6.41	17.04	0.51	6.08	2.81	52.83
	Centerpoint [34]	0.07	0.30	23.01	0.02	5.85	0.44	4.55	30.74	0.33	9.01	1.17	15.98
	PVRCNN [17]	22.20	29.12	52.74	6.06	27.53	30.02	37.64	55.52	10.07	33.31	12.42	54.05
	Second [15]	19.23	38.89	42.47	14.83	28.85	35.78	42.55	50.18	19.34	36.96	4.63	18.19
	Pointpillar [27]	24.90	48.10	55.20	16.58	36.20	41.06	52.51	61.63	24.1	44.82	4.85	14.49
	Tanet [35]	25.14	49.66	53.54	12.48	35.20	44.06	51.55	55.24	21.25	43.03	6.52	23.45
	CT3D [36]	9.09	7.92	10.06	11.62	9.67	10.06	14.92	14.65	14.05	14.39	6.11	49.87
Multi frame	Motionnet [24] *	29.30	48.80	59.38	19.96	39.36	**47.44**	51.99	63.96	25.05	47.11	5.51	40.31
	PP-REC [37]	25.43	**49.78**	53.74	17.00	36.49	37.50	**52.72**	59.54	24.02	43.45	6.48	75.26
	Pointpillar ($[x, y, z, v, p, t]$) [b]	27.02	43.84	65.54	15.20	37.90	36.79	46.75	67.33	20.85	43.93	4.85	14.52
	RadarMFNet (this study)	**29.36**	49.23	**71.22**	**20.62**	**42.61**	44.66	51.11	**74.25**	**26.27**	**49.07**	6.06	40.00

The targets with zero points in the ground truth box are filtered out during the precision calculation
* A modified Motionet is used for comparison, which also uses the pillar method to encode the point cloud.
[b] A time dimension was added to input point clouds of different frames into the network.
[d] The inference times are calculated using NVIDIA TITAN RTX graphics card.

TABLE III
IMPACT OF DIFFERENT ST BLOCKS ON 3-D AP FOR DIFFERENT OBJECT TYPES

Method	Car	Pedestrian	Cyclist	Truck	mAp
ST block A	28.44	52.25	64.44	17.89	40.76
ST block B	29.26	46.93	65.51	21.37	40.77
ST block C	28.40	45.21	63.15	20.95	39.43
This study	29.36	49.23	71.22	20.62	42.61

the model. In Fig. 10(d) and (e), the fit of the target contour is also limited, owing to fewer target points.

C. Quantitative Analysis

The 3-D and BEV-detection results of our method and other point-cloud-detection networks are shown in Table II. The detection accuracy AP and mean accuracy mAP for each class are shown in the detection results, where bold font indicates the best performance.

As shown in Table II, the framework proposed in this study achieves good detection performance on both the 3-D-detection and BEV-detection benchmarks, with mAPs of 42.61 and 49.07, respectively. For 2-D BEV detection, the proposed method achieved the best results for cyclists and trucks.

It can be observed that the detection accuracy for pedestrians and cyclists is higher, whereas that for cars and trucks is lower compared with the results of other networks. This occurs due to the difference in the size of different targets. The 3-D detection IOU values of other 3-D object-detection algorithms for pedestrians and cyclists, as well as cars and trucks, are 0.25 and 0.5, respectively. In addition, owing to the sparseness of millimeter-wave radar point-cloud targets, the contours of car and truck targets cannot be fully reflected; therefore, the detection accuracy of our proposed method is lower. Another reason is due to the large number of static vehicles and truck targets marked in the dataset. These targets have a Doppler velocity of 0, thereby making it difficult to distinguish them from other static targets on both sides of the road.

In comparison with other object-detection methods, methods such as 3DSSD, which is based on the direct processing of a raw point cloud, and Centerpoint, which is based on the anchor-free method, have not achieved good results in sparse 4-D radar point-cloud detection. By analyzing the intermediate process of network training, it is observed that the method of using a voxel grid to encode a sparse point cloud and the anchor-based method are better able to converge the network.

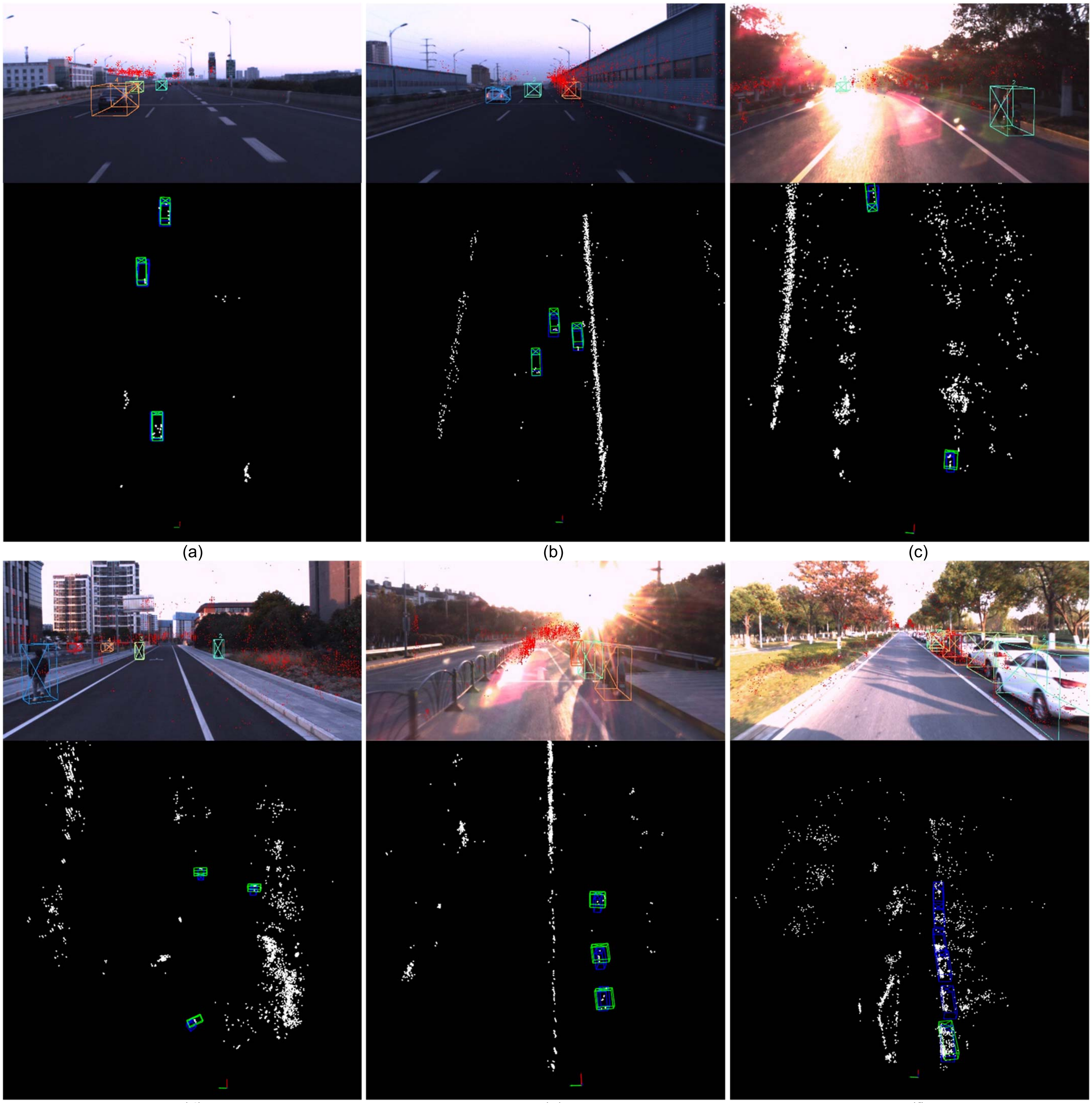
This study designed a spatiotemporal extraction module ST block for the spatiotemporal feature extraction of the backbone network, thereby improving detection accuracy.

Fig. 11 shows the detection effects of different targets under different IOU conditions. It can be observed from Fig. 12 that the detection performances for pedestrians, cyclists, and trucks decrease with an increase in the target distance, and that for cars fluctuates with an increase in distance. This may occur because the radar detection of dynamic vehicle targets is more stable in the middle distance of the millimeter-wave radar.

VI. ABLATION STUDY

A. ST Blocks With Different Structures

As shown in Fig. 13, different ST blocks are designed and compared in the experiments. The ST block is used for the



(a) (b) (c)

(d) (e) (f)

Fig. 10. Display of some network-detection experimental results. The 3-D detection result in the picture is the true value, the blue box in the point-cloud detection result is the 3-D true value box, and the green box is the predicted value. (a) Detection of vehicle objects in an elevated road scene. (b) Detection of vehicles in a road scene with soundproof strips. (c) Detection of motorcycles in a road scene with green belts. (d) Detection of pedestrians in a road scene. (e) Detection of motorcycles in a road scene with a fence. (f) Detection of a static vehicle object in a scene with a green belt.

extraction of spatiotemporal features from the top-down network in the backbone network. The first layer of the ST block module is used for scale reduction and channel increase. The first layer and subsequent convolution layers of ST block A use $3 \times 3 \times 3$ convolution kernels. The first layer in ST block B uses a $3 \times 3 \times 3$ convolution kernel, and the subsequent layers use a $1 \times 3 \times 3$ convolution kernel. The subsequent layers do not extract temporal features but only spatial features. The first layer in ST block C uses $3 \times 1 \times 1$ and $1 \times 3 \times 3$ convolution kernels for feature extraction consecutively, and the subsequent layers use $1 \times 3 \times 3$ convolution kernels for feature extraction.

The results in Table III show that, in the ST block, the subsequent layers can use $1 \times 3 \times 3$ convolution kernels to

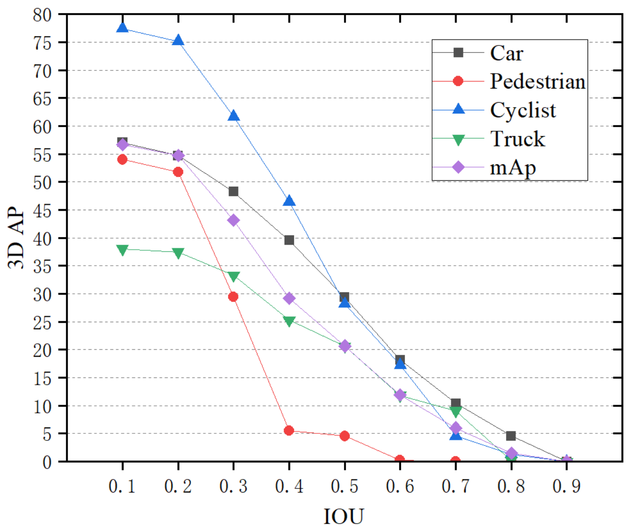


Fig. 11. Impact of different 3-D IOUs on 3-D AP for different object types.

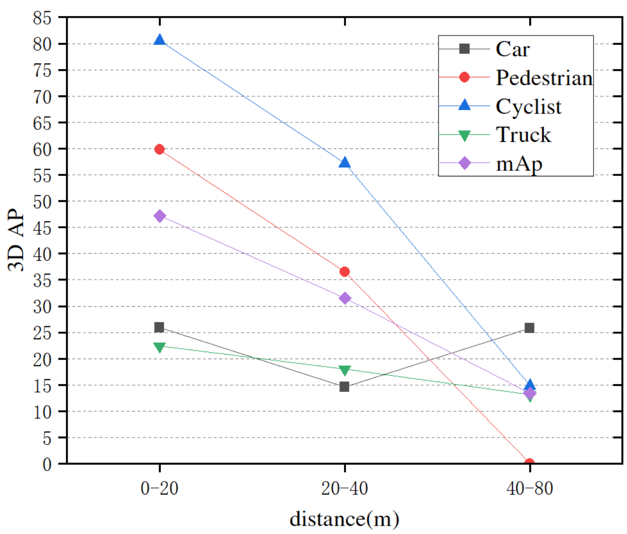


Fig. 12. Impact of different distances on 3-D AP for different object types.

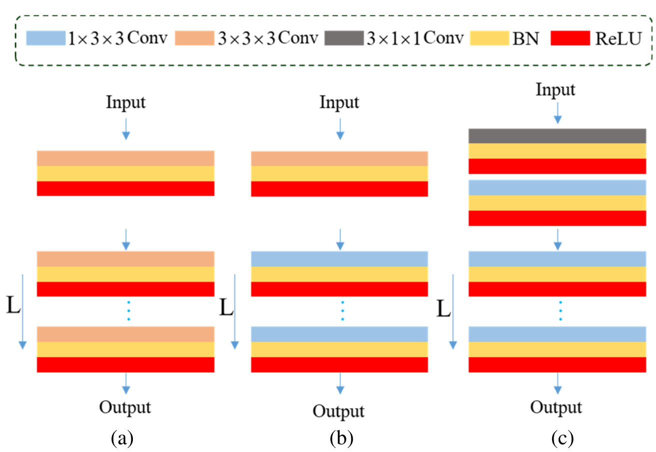


Fig. 13. Three differently designed ST blocks for comparison. (a) ST block A. (b) ST block B. (c) ST block C.

reduce the number of parameters, and using 3 × 3 × 3 convolution kernels in the first layer enables better results. Moreover, a 1 × 3 × 3 convolution is added to the first layer in the model, which can strengthen the network's extraction of spatial features, thereby producing better results.

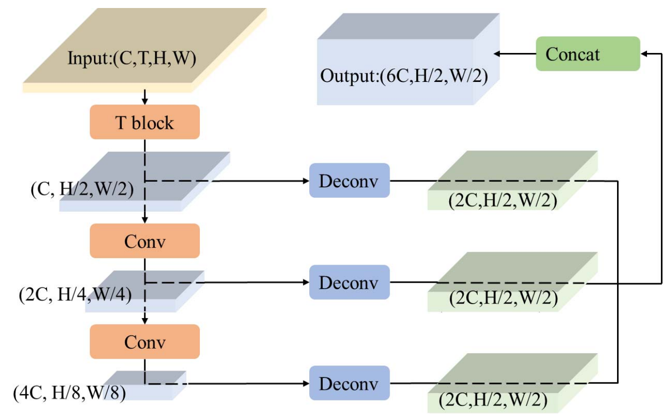


Fig. 14. Backbone network for different types of spatiotemporal feature extraction. It includes a T block that comprises 3-D convolution for spatiotemporal extraction, a 2-D top-down feature extraction network, and an upsampling layer for multiscale feature fusion.

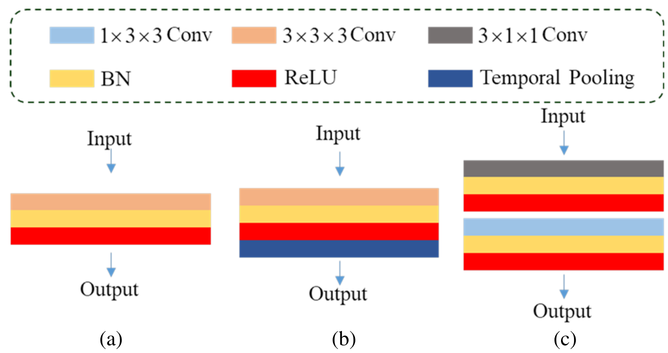


Fig. 15. Three differently designed T blocks for comparison. (a) T block A. (b) T block B. (c) T block C.

B. Multiframe-Detection Backbone With Different Structures

As shown in Figs. 14 and 15, different simplified networks are designed for comparison. The network utilized for comparison uses a T block module in Fig. 15 to extract the 3-D temporal and spatial features of the input point cloud. The latter network uses a method that is similar to the backbone of the pointpillar [27] network for feature extraction. A top-down network performs spatial feature extraction, and an upsampling layer performs multiscale feature fusion. Three different T block modules are designed for comparison. T block A uses a 3 × 3 × 3 convolution kernel and downsamples the temporal dimension to one dimension on the convolutional layer. T block B uses a 3 × 3 × 3 convolution kernel and uses maxpooling to downsample the temporal dimension. T block C uses 3 × 1 × 1 and 1 × 3 × 3 convolution kernels in Motionnet [35] and downsamples the temporal dimension to one dimension in the convolution.

The experimental results show that T block A achieves good results, whereas the results of T block B are unstable, and those of T block C are poor. The results of the network that includes

TABLE IV
IMPACT OF DIFFERENT T BLOCKS ON 3-D AP FOR DIFFERENT OBJECT TYPES

Method	Car	Pedestrian	Cyclist	Truck	mAp
T block A	24.96	51.62	63.75	22.26	40.65
T block B	29.92	41.87	64.74	17.46	38.50
T block C	23.74	48.90	54.79	17.09	36.13

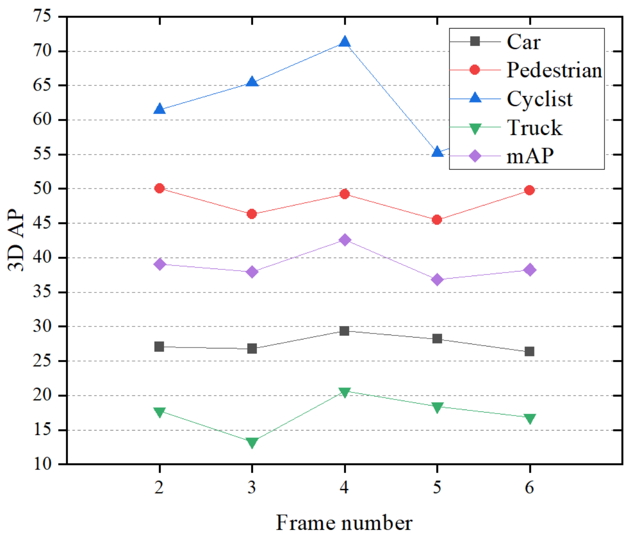


Fig. 16. Impact of different frame numbers on 3-D AP for different object types.

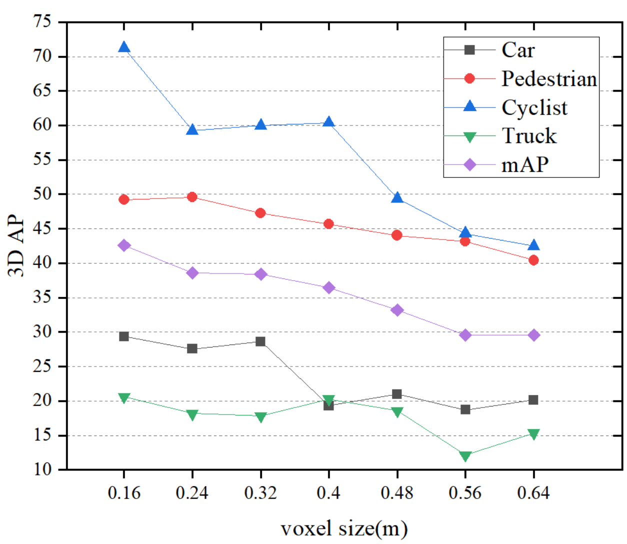


Fig. 17. Impact of different voxel sizes on 3-D AP for different object types.

the T block A module are better, the model size is smaller, and the effect of this network is better than directly inputting the temporal dimension to the single-frame network, as shown in Table IV. Thus, this network can be used as a simplified multiframe point-cloud extraction network.

TABLE V
IMPACT OF MOTION COMPENSATION ON 3-D AP FOR DIFFERENT OBJECT TYPES

Method	Car	Pedestrian	Cyclist	Truck	mAp
No compensation	20.71	41.71	66.86	17.83	36.78
With compensation	29.36	49.23	71.22	20.62	42.61

C. Number of Frames

The experimental results for the detection performance of the network within two to six frames are shown in Fig. 16. The network that uses four frames achieves the best 3-D mAp results and the best 3-D AP results in the car, cyclist, and truck categories. The network that uses two frames achieves the best 3-D AP results in the pedestrian category.

D. Voxel Size

The experimental results for the voxel size of the pillar in the network are shown in Fig. 17. The network with a grid of 0.16 m achieved the best 3-D mAP results. Smaller grids within the range improve the comprehensive object-detection performance. The parameter quantity of the network model increases by four times when the grid size is considered halved; therefore, the selection of the existing model grid comprehensively considers accuracy and speed.

E. Self-Motion Compensation

The experimental results on implementing motion compensation when inputting multiframe point clouds are shown in Table V. The results show that motion compensation can improve network performance.

VII. CONCLUSION

This study proposes a framework based on multiframe 4-D radar point clouds for multiclass object detection. Improving the performance of the single-frame millimeter-wave radar-detection method is limited, owing to the sparseness of the 4-D radar point cloud. The multiframe millimeter-wave radar point cloud is used as the input, and the data preprocessing step only inputs millimeter-wave radar point clouds, but no other sensor data. Most of the existing 3-D object-detection methods are based on LiDAR; however, the detection effect of these methods on the 4-D radar is limited. This study compares different single-frame- and multiframe-based 3-D object-detection methods, and our proposed method achieved the best results. The experimental results show that, in an actual complex road environment, the 4-D millimeter-wave radar point cloud has a better detection effect on pedestrian and cyclist objects; however, the detection effect on car and truck objects is still limited.

ACKNOWLEDGMENT

The TJ4DRadSet [11] dataset will be released at https://github.com/TJRadarLab/TJ4DRadSet.

References

[1] Q. Chen, Y. Xie, S. Guo, J. Bai, and Q. Shu, "Sensing system of environmental perception technologies for driverless vehicle: A review of state of the art and challenges," *Sens. Actuators A, Phys.*, vol. 319, Mar. 2021, Art. no. 112566.

[2] O. Schumann, C. Wohler, M. Hahn, and J. Dickmann, "Comparison of random forest and long short-term memory network performances in classification tasks using radar," in *Proc. Sensor Data Fusion: Trends, Solutions, Appl. (SDF)*, Oct. 2017, pp.‘1–6.

[3] J. Lombacher, M. Hahn, J. Dickmann, and C. Wohler, "Object classification in radar using ensemble methods," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Mar. 2017, pp. 87–90.

[4] N. Scheiner, N. Appenrodt, J. Dickmann, and B. Sick, "Radar-based feature design and multiclass classification for road user recognition," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2018, pp. 779–786.

[5] A. Danzer, T. Griebel, M. Bach, and K. Dietmayer, "2D car detection in radar data with pointnets," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, Oct. 2019, pp. 61–66.

[6] O. Schumann, M. Hahn, J. Dickmann, and C. Wohler, "Semantic segmentation on radar point clouds," in *Proc. 21st Int. Conf. Inf. Fusion (FUSION)*, Jul. 2018, pp. 2179–2186.

[7] M. Meyer and G. Kuschk, "Automotive radar dataset for deep learning based 3D object detection," in *Proc. 16th Eur. Radar Conf. (EuRAD)*, Oct. 2019, pp. 129–132.

[8] A. Palffy, E. Pool, S. Baratam, J. F. P. Kooij, and D. M. Gavrila, "Multi-class road user detection with 3+1D radar in the view-of-delft dataset," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 4961–4968, Apr. 2022.

[9] M. Meyer and G. Kuschk, "Deep learning based 3D object detection for automotive radar and camera," in *Proc. 16th Eur. Radar Conf. (EuRAD)*, Oct. 2019, pp. 133–136.

[10] B. Xu, "RPFA-Net: A 4D RaDAR pillar feature attention network for 3D object detection," in *Proc. IEEE Int. Intell. Transp. Syst. Conf. (ITSC)*, Sep. 2021, pp. 3061–3066.

[11] L. Zheng et al., "TJ4DRadSet: A 4D radar dataset for autonomous driving," 2022, *arXiv:2204.13483*.

[12] L.-H. Wen and K.-H. Jo, "Deep learning-based perception systems for autonomous driving: A comprehensive survey," *Neurocomputing*, vol. 489, pp. 255–270, Jun. 2022.

[13] D. Fernandes, "Point-cloud based 3D object detection and classification methods for self-driving applications: A survey and taxonomy," *Inf. Fusion*, vol. 68, pp. 161–191, Apr. 2021.

[14] Y. Zhou and O. Tuzel, "VoxelNet: End-to-end learning for point cloud based 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4490–4499.

[15] Y. Yan, Y. Mao, and B. Li, "SECOND: Sparsely embedded convolutional detection," *Sensors*, vol. 18, no. 10, p. 3337, Oct. 2018.

[16] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, "PointPillars: Fast encoders for object detection from point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12689–12697.

[17] S. Shi et al., "PV-RCNN: Point-voxel feature set abstraction for 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 10529–10538.

[18] Z. Yang, Y. Sun, S. Liu, and J. Jia, "3DSSD: Point-based 3D single stage object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11040–11048.

[19] S. Shi, X. Wang, and H. Li, "PointRCNN: 3D object proposal generation and detection from point cloud," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 770–779.

[20] Z. Yang, Y. Sun, S. Liu, X. Shen, and J. Jia, "STD: Sparse-to-dense 3D object detector for point cloud," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1951–1960.

[21] R. Huang et al., "An LSTM approach to temporal 3D object detection in LiDAR point clouds," 2020, *arXiv:2007.12392*.

[22] C. Tu, E. Takeuchi, A. Carballo, and K. Takeda, "Point cloud compression for 3D LiDAR sensor using recurrent neural network with residual blocks," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2019, pp. 3274–3280.

[23] J. Yin, J. Shen, C. Guan, D. Zhou, and R. Yang, "LiDar-based online 3D video object detection with graph-based message passing and spatiotemporal transformer attention," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Mar. 2020, pp. 11495–11504.

[24] P. Wu, S. Chen, and D. N. Metaxas, "MotionNet: Joint perception and motion prediction for autonomous driving based on bird's eye view maps," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 11385–11395.

[25] J. Lombacher, M. Hahn, J. Dickmann, and C. Wohler, "Potential of radar for static object classification using deep learning methods," in *IEEE MTT-S Int. Microw. Symp. Dig.*, May 2016, pp. 1–4.

[26] R. Prophet, A. Deligiannis, J.-C. Fuentes-Michel, I. Weber, and M. Vossiek, "Semantic segmentation on 3D occupancy grids for automotive radar," *IEEE Access*, vol. 8, pp. 197917–197930, 2020.

[27] D. Kellner, M. Barjenbruch, J. Klappstein, J. Dickmann, and K. Dietmayer, "Instantaneous ego-motion estimation using Doppler radar," in *Proc. 16th Int. IEEE Conf. Intell. Transp. Syst. (ITSC)*, Oct. 2013, pp. 869–874.

[28] Y. Almalioglu, M. Turan, C. X. Lu, N. Trigoni, and A. Markham, "Milli-RIO: Ego-motion estimation with low-cost millimetre-wave radar," *IEEE Sensors J.*, vol. 21, no. 3, pp. 3314–3323, Feb. 2021.

[29] M. Holder, S. Hellwig, and H. Winner, "Real-time pose graph SLAM based on radar," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2019, pp. 1145–1151.

[30] F. Schuster, C. G. Keller, M. Rapp, M. Haueis, and C. Curio, "Landmark based radar SLAM using graph optimization," in *Proc. IEEE 19th Int. Conf. Intell. Transp. Syst. (ITSC)*, Nov. 2016, pp. 2559–2564.

[31] B. Yang, R. Guo, M. Liang, S. Casas, and R. Urtasun, "RadarNet: Exploiting radar for robust perception of dynamic objects," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 496–512.

[32] K. G. Derpanis, "Overview of the RANSAC algorithm," Image Rochester NY, May 2010, pp. 2–3, vol. 4, no. 1.

[33] S. Rusinkiewicz and M. Levoy, "Efficient variants of the ICP algorithm," in *Proc. 3rd Int. Conf. 3-D Digit. Imag. Modeling*, May/Jun. 2001, pp. 145–152.

[34] T. Yin, X. Zhou, and P. Krahenbuhl, "Center-based 3D object detection and tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 11784–11793.

[35] Z. Liu, X. Zhao, T. Huang, R. Hu, Y. Zhou, and X. Bai, "TANet: Robust 3D object detection from point clouds with triple attention," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 11677–11684.

[36] H. Shenga, "Improving 3D object detection with channel-wise transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 2743–2752.

[37] S. McCrae and A. Zakhor, "3D object detection for autonomous driving using temporal LiDAR data," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 2661–2665.

Bin Tan received the B.S. degree from Central South University, Changsha, China, in 2016. He is currently pursuing the Ph.D. degree in vehicle engineering with the School of Automotive Studies, Tongji University, Shanghai, China.

His research interests include millimeter-wave radar signal processing, deep learning research on millimeter-wave radar point-cloud data and RF signal data, object detection and tracking, and multisensor fusion.

Zhixiong Ma is a Lecturer with the School of Automotive Science and Technology, Tongji University, Shanghai, China. He has presided over several corporate and government projects. He has published a series of academic papers in domestic and international academic journals and has obtained a total of several invention patents. His main research interests are automotive safety. His research interests include automotive active and passive safety technologies, and autonomous vehicle perception systems.

Dr. Ma won the Third Prize for Science and Technology Progress of China Automobile Industry in 2005 and the Third Prize of Shanghai Science and Technology Progress in 2009.

Xichan Zhu received the Ph.D. degree from the Department of Automotive Engineering, Tsinghua University, Beijing, China, in 1995.

He is a Professor and a Doctoral Supervisor with the School of Automotive Engineering, Tongji University, Shanghai, China. He presided over and completed a number of enterprise and government projects. He has published many academic papers in domestic and foreign academic journals. His research interests include the active and passive safety of automobiles.

Dr. Zhu won the First Prize for Scientific and Technological Progress of the Ministry of Machinery Industry in 1996 and the First Prize for Automobile Industry Scientific and Technological Progress. In 2001, he won the Second Prize in the Automobile Industry's Scientific and Technological Progress. In 2004, he won the Second Prize in the Automobile Industry's Scientific and Technological Progress. He is the Vice-Chairperson of the Safety Branch of the Chinese Society of Automotive Engineers.

Sen Li received the B.S. degree in automotive engineering from Southwest Jiaotong University, Chengdu, Sichuan, China, in 2016. He is currently pursuing the Ph.D. degree with the School of Automotive Studies and the Intelligent Vehicle Research Institute, Tongji University, Shanghai, China.

His current research interests include multisensor data fusion, multiobject detection and tracking, deep learning, and intelligent driving technology.

Lianqing Zheng received the B.S. degree in vehicle engineering from the Harbin Institute of Technology, Weihai, China, in 2019. He is currently pursuing the Ph.D. degree in vehicle engineering with the School of Automotive Studies, Tongji University, Shanghai, China.

His research interests include image processing and analysis, object detection, multisensor fusion, radar signal processing, and deep learning.

Sihan Chen, photograph and biography not available at the time of publication.

Libo Huang received the B.S. degree in radio engineering from Southeast University, Nanjing, China, in 2002, and the M.S. and Ph.D. degrees in electronics and information technology from the Technical University of Munich, Munich, Germany, in 2004 and 2008, respectively.

He is currently an Associate Professor with the School of Information and Electrical Engineering, Zhejiang University City College, Hangzhou, China. His research interests include automobile environment perception, information fusion, automotive radar antenna design, and digital signal processing.

Jie Bai received the B.S. degree in instrument science and technology from the Harbin Institute of Technology, Harbin, China, in 1987, and the M.S. and Ph.D. degrees in mechanical engineering from Hiroshima University, Higashihiroshima, Japan, in 1992 and 1995, respectively.

He is currently a Professor with the School of Information and Electrical Engineering, Zhejiang University City College, Hangzhou, China. He is also the Secretary-General of the Intelligent Transportation Branch of the China Automotive Engineering Institute, Tianjin, China. His current research interests include intelligent automotive environment perception and decision-making, multisensor data fusion, deep learning, and signal processing.

Dr. Bai is the Vice-Chairperson of the SAE International Conference Organizing Committee and the Chairperson of the Technical Expert Committee.